



NCI Protocol #: 10323

Patient ID:

Date:

1 of 20

Patient Name:

Date Collected:

Biopsy Site: **Bladder**

Sample ID:

Telephone:

Tumor Content (%): **20_49**

Patient DOB:

Referring Physician:

Primary Diagnosis: **Papillary urothelial carcinoma**

MoCha ID:

Fax:

Preanalytics to include tumor enrichment as required is performed by Van Andel Research Institute; Grand Rapids; MI 49503

Table of Contents

Page

Variant Details

2

Biomarker Descriptions

3

Relevant Therapy Summary

7

Clinical Trials Summary

10

Report Highlights

8 Relevant Biomarkers

3 Therapies Available

43 Clinical Trials

Relevant Bladder Urothelial Carcinoma Findings

Gene	Finding
FGFR2	None detected
FGFR3	None detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
III	PALB2 G1135A	None	olaparib ¹ rucaparib	17
III	BRAF R671Q	None	trametinib	18
III	ARID1A Q581*	None	None	9
III	SMARCA4 P22R	None	None	4
IIC	TP53 R248W	None	None	3
III	SETD2 S188*	None	None	2
III	PTCH1 Q417E	None	None	2

Public data sources included in relevant therapies: FDA¹, NCCN

Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

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Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2023.06(006). The content of this report has not been evaluated or approved by FDA, EMA or other regulatory agencies.

Relevant Biomarkers (continued)

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
III	TSC2 E588Q	None	None	1

Public data sources included in relevant therapies: FDA1, NCCN
Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

Prevalent cancer biomarkers without relevant evidence based on included data sources

AXL E682M, GATA2 L305V

Variant Details

DNA Sequence Variants								
Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Coverage	Transcript	Variant Effect
ARID1A	p.(Q581*)	c.1741C>T	.	chr1:27058033	5.91%	812	NM_006015.4	nonsense
SETD2	p.(S188*)	c.563C>G	.	chr3:47165563	5.11%	1996	NM_014159.6	nonsense
GATA2	p.(L305V)	c.913C>G	.	chr3:128202807	19.45%	1208	NM_032638.4	missense
BRAF	p.(R671Q)	c.2012G>A	.	chr7:140439727	20.53%	794	NM_004333.4	missense
PTCH1	p.(Q417E)	c.1249C>G	.	chr9:98240435	5.17%	1430	NM_000264.3	missense
TSC2	p.(E588Q)	c.1762G>C	.	chr16:2120502	13.70%	657	NM_000548.3	missense
PALB2	p.(G1135A)	c.3404G>C	.	chr16:23614937	28.41%	602	NM_024675.3	missense
TP53	p.(R248W)	c.742C>T	COSM10656	chr17:7577539	53.39%	869	NM_000546.5	missense
SMARCA4	p.(P22R)	c.65C>G	.	chr19:11094892	27.76%	281	NM_001128849.1	missense
AXL	p.(E682M)	c.2044_2045delGAin sAT	.	chr19:41762364	7.52%	612	NM_021913.4	missense

Comments:

NA

Report Signed By

Name	Role	Date

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

Biomarker Descriptions

ARID1A (AT-rich interaction domain 1A)

Background: The ARID1A gene encodes the AT-rich interaction domain 1A tumor suppressor protein¹. ARID1A, also known as BAF250A, belongs to the ARID1 subfamily that also includes ARID1B^{1,2}. ARID1A and ARID1B are mutually exclusive subunits of the BAF variant of the SWI/SNF chromatin-remodeling complex^{2,3}. The BAF complex is a multisubunit protein that consists of SMARCB1/IN1, SMARCC1/BAF155, SMARCC2/BAF170, SMARCA4/BRG1 or SMARCA2/BRM, and ARID1A or ARID1B³. The BAF complex remodels chromatin at promoter and enhancer elements to alter and regulate gene expression^{3,4}. ARID1A binds to transcription factors and coactivator/corepressor complexes to alter transcription². Recurrent inactivating mutations in BAF complex subunits, including ARID1A, lead to transcriptional dysfunction thereby, altering its tumor suppressor function².

Alterations and prevalence: Mutations in SWI/SNF complex subunits are the most commonly mutated chromatin modulators in cancer and have been observed in 20% of all tumors⁴. The majority of ARID1A inactivating mutations are nonsense or frameshift mutations². Somatic mutations in ARID1A have been identified in 50% of ovarian clear cell carcinoma, 30% of endometrioid carcinoma, and 24-43% of uterine corpus endometrial carcinoma, bladder urothelial carcinoma, and stomach adenocarcinoma^{3,5,6}. In microsatellite stable (MSS) colorectal cancer, mutations in ARID1A have been observed to correlate with increased tumor mutational burden (TMB) and expression of genes involved in the immune response⁷.

Potential relevance: Currently, no therapies are approved for ARID1A aberrations. However, the FDA has granted fast track designation to HSF1 pathway inhibitor, NXP8008, for the treatment of platinum resistant ARID1A-mutated ovarian carcinoma.

AXL (AXL receptor tyrosine kinase)

Background: The AXL gene encodes the AXL receptor tyrosine kinase (RTK), a member of the TYRO3/AXL/MERTK (TAM) RTK subfamily. AXL is activated by binding to the growth arrest specific protein 6 (GAS6) ligand as part of the AXL/Gas6 pathway which regulates endothelial cell survival and angiogenesis^{9,10}. AXL activates downstream signaling of the RAS/RAF/MEK/ERK, PI3K/AKT/mTOR, NF-κB, and JAK/STAT pathways thereby promoting tumor cell proliferation, migration, and invasion^{9,11,12,13}.

Alterations and prevalence: Mutations in AXL are found in 8% of uterine cancer and melanoma, and 4-6% of stomach cancer^{5,6,14,15}. Amplification of AXL is found to occur in up to 9% of uterine cancer, 4% of bladder cancer, and 3% of ovarian cancer^{5,16}. Upregulation due to overexpression is observed in a variety of tumor types, although overexpression is not correlated with genomic alterations such as amplifications or mutations⁹.

Potential relevance: Currently, no therapies are approved for AXL aberrations, although cabozantinib (2012), an FDA approved multi-targeted tyrosine kinase inhibitor, includes AXL as a target. AXL overexpression confers poor prognosis and poor survival in various cancers including glioblastoma multiforme, mesothelioma, breast, pancreatic, renal cell, and esophageal cancers^{17,18,19,20,21,22}. Importantly, upregulation of AXL is an alternative acquired resistance mechanism that confers resistance to EGFR targeted therapies including erlotinib and cetuximab in non-small cell lung cancer^{23,24}. In a phase I clinical trial, a single patient with AXL amplification demonstrated partial response when treated with the investigational MET/AXL inhibitor glesatinib²⁵.

BRAF (B-Raf proto-oncogene, serine/threonine kinase)

Background: The BRAF gene encodes the B-Raf proto-oncogene serine/threonine kinase, a member of the RAF family of serine/threonine protein kinases which also includes ARAF and RAF1 (CRAF). BRAF is among the most commonly mutated kinases in cancer. Activation of the MAPK pathway occurs through BRAF mutations and leads to an increase in cell division, dedifferentiation, and survival^{26,27}. BRAF mutations are categorized into three distinct functional classes namely, class 1, 2, and 3, and are defined by the dependency on the RAS pathway. Class 1 and 2 BRAF mutants are RAS-independent in that they signal as active monomers (Class 1) or dimers (Class 2) and become uncoupled from RAS GTPase signaling, resulting in constitutive activation of BRAF²⁸. Class 3 mutants are RAS dependent as the kinase domain function is impaired or dead^{28,29,30}.

Alterations and prevalence: Recurrent somatic mutations in BRAF are observed in 40-60% of melanoma and thyroid cancer, approximately 10% of colorectal cancer, and about 2% of non-small cell lung cancer (NSCLC)^{5,6,31,32,33}. Mutations at V600 belong to class 1 and include V600E, the most recurrent somatic BRAF mutation across diverse cancer types^{29,34}. Class 2 mutations include K601E/N/T, L597Q/V, G469A/V/R/, G464V/E/, and BRAF fusions²⁹. Class 3 mutations include D287H, V459L, G466V/E/A, S467L, G469E, and N581S/I²⁹. BRAF V600E is universally present in hairy cell leukemia, mature B-cell cancer, and prevalent in histiocytic neoplasms^{35,36,37}. Other recurrent BRAF somatic mutations cluster in the glycine-rich phosphate-binding loop at codons 464-469 in

Biomarker Descriptions (continued)

exon 11 as well as additional codons flanking V600 in the activation loop³⁴. In primary cancers, BRAF amplification is observed in 8% of ovarian cancer and about 1% of breast cancer^{5,6}. BRAF fusions are mutually exclusive to BRAF V600 mutations and have been described in melanoma, thyroid cancer, pilocytic astrocytoma, NSCLC, and several other cancer types^{38,39,40,41,42}. Part of the oncogenic mechanism of BRAF gene fusions is the removal of the N-terminal auto-inhibitory domain leading to constitutive kinase activation^{30,38,40}.

Potential relevance: Vemurafenib⁴³ (2011) was the first targeted therapy approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E mutation. BRAF class 1 mutations, including V600E, are sensitive to vemurafenib, whereas class 2 and 3 mutations are insensitive²⁹. BRAF kinase inhibitors including dabrafenib⁴⁴ (2013) and encorafenib⁴⁵ (2018) are also approved for the treatment of patients with unresectable or metastatic melanoma with BRAF V600E/K mutations. Encorafenib⁴⁵ is approved in combination with cetuximab⁴⁶ (2020) for the treatment of BRAF V600E mutated colorectal cancer. Due to the tight coupling of RAF and MEK signaling, several MEK inhibitors have been approved for patients harboring BRAF alterations²⁹. Trametinib⁴⁷ (2013) and binimetinib⁴⁸ (2018) were approved for the treatment of metastatic melanoma with BRAF V600E/K mutations. Combination therapies of BRAF plus MEK inhibitors have been approved in melanoma and NSCLC. The combinations of dabrafenib/trametinib (2015) and vemurafenib/cobimetinib⁴⁹ (2015) were approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E/K mutation. Subsequently, the combination of dabrafenib and trametinib was approved for metastatic NSCLC (2017) with a BRAF V600E mutation. The PD-L1 antibody, atezolizumab⁵⁰, has also been approved in combination with cobimetinib and vemurafenib for BRAF V600 mutation-positive unresectable or metastatic melanoma. In 2018, binimetinib⁵¹ was also granted breakthrough designation in combination with cetuximab and encorafenib for BRAF V600E mutant metastatic colorectal cancer. The ERK inhibitor ulixertinib⁵² was granted fast track designation in 2020 for the treatment of patients with non-colorectal solid tumors harboring BRAF mutations G469A/V, L485W, or L597Q. The FDA granted fast track designation (2022) to the pan-RAF inhibitor, KIN-2787⁵³, for the treatment of BRAF class II or III alteration-positive malignant or unresectable melanoma. BRAF fusion is a suggested mechanism of resistance to BRAF targeted therapy in melanoma⁵⁴. Additional mechanisms of resistance to BRAF targeted therapy include BRAF amplification and alternative splice transcripts as well as activation of PI3K signaling and activating mutations in KRAS, NRAS, and MAP2K1/2 (MEK1/2)^{55,56,57,58,59,60,61}. Clinical responses to sorafenib and trametinib in limited case studies of patients with BRAF fusions have been reported⁴².

GATA2 (GATA binding protein 2)

Background: The GATA2 gene encodes the GATA binding protein 2, a zinc-finger transcription factor responsible for regulating transcription of many genes by binding to the DNA consensus sequence T/A(GATA)A/G^{62,63}. GATA2 is highly expressed in hematopoietic cells and is a critical component for the proliferation and maintenance of stem cells⁶². Germline mutations are spontaneous and are inherited in an autosomal dominant pattern. Germline mutations predispose patients to myelodysplastic syndrome and/or acute myeloid leukemia (AML)^{64,65}. GATA2 deficiency is recognized as a myelodysplastic syndrome (MDS) predisposition syndrome⁶².

Alterations and prevalence: GATA2 somatic mutations cause loss of function in the mutated allele which leads to haploinsufficiency⁶⁶. Hemizygous mutations in GATA2 frequently co-occur with the inv(3)/t(3;3) aberration that involves the fusion of RPN1/MECOM⁶⁷. Up to 10% of intermediate-risk karyotype AML with CEBPA mutations also harbor GATA2 somatic mutations⁶². The gain-of-function mutation GATA2 L359V has been identified in 10% of chronic myeloid leukemia during blast phase⁶⁸. Amplifications in GATA2 are observed in up to 6% of lung squamous cell carcinoma and cervical squamous cell carcinoma, and up to 3% of ovarian serous cystadenocarcinoma and esophageal adenocarcinoma^{5,6}.

Potential relevance: GATA2 nonsense, frameshift, or splice site missense mutations in codons 349-398 confer poor prognosis in MDS⁶⁹. GATA2-MECOM fusion is associated with adverse risk in acute myeloid leukemia (AML)⁷⁰. GATA2-MECOM fusion with FLT3 ITD or FLT3 TKD mutation are indicated for combination therapy with midostaurin, cytarabine and daunorubicin in acute myeloid leukemia⁷⁰.

PALB2 (partner and localizer of BRCA2)

Background: The PALB2 gene encodes the partner and localizer of BRCA2 protein that binds to and promotes intranuclear localization of the breast cancer 2 early onset (BRCA2) protein⁷¹. Also known as FANCN, PALB2 belongs to the Fanconi Anemia (FA) complementation group of proteins that also include FANCA, FANCB, FANCC, FANCD1 (BRCA2), FANCD2, FANCE, FANCF, FANCG, FANCI, FANCL (BRIP1), FANCL, and FANCM. FA genes are tumor suppressors that play a role in interstrand cross-link (ICL) DNA repair through homologous recombination repair (HRR) of double-strand breaks (DSB) and nucleotide excision repair (NER)⁷². Loss of function mutations of genes in the FA family and HRR pathway, including PALB2, can result in the BRCAness phenotype, characterized by a

Biomarker Descriptions (continued)

defect in the HRR pathway, mimicking BRCA1 or BRCA2 loss^{73,74}. Germline mutations in FA genes lead to Fanconi Anemia, a condition characterized by chromosomal instability and congenital abnormalities including bone marrow failure and cancer predisposition^{75,76}. Specifically, biallelic germline mutations resulting in PALB2 loss of function confer a predisposition to pediatric malignancies^{77,78}. Additionally, monoallelic germline mutations in PALB2 have been associated with an increased risk of developing breast cancer^{77,79}.

Alterations and prevalence: Somatic alterations in PALB2 include missense or truncating mutations and are observed in 2-6% of melanoma, uterine, bladder, breast, lung, stomach and colorectal cancers⁵.

Potential relevance: The PARP inhibitor, olaparib⁸⁰ is approved (2020) for metastatic castration-resistant prostate cancer (mCRPC) with deleterious or suspected deleterious, germline or somatic mutations in HRR genes that includes PALB2. In a phase II trial of patients with metastatic, castration-resistant prostate cancer, one patient exhibiting a somatic PALB2 frameshift mutation exhibited durable response to olaparib for 39 weeks^{81,82}. However, olaparib resistance was observed following 9-months of treatment due to the emergence of a secondary deletion which restored the PALB2 reading frame, a resistance mechanism similar to that observed in PARPi treated BRCA mutated patients^{82,83}. In 2022, the FDA granted fast track designation to the small molecule inhibitor, pidnarulex⁸⁴, for BRCA1/2, PALB2, or other homologous recombination deficiency (HRD) mutations in breast and ovarian cancers.

PTCH1 (patched 1)

Background: The PTCH1 gene encodes the patched 1 protein, a transmembrane protein that along with PTCH2, belongs to the patched gene family¹. PTCH1 is involved in the Hedgehog (Hh) signaling pathway that plays a significant role in embryonic development, cell proliferation, and cell differentiation^{85,86}. PTCH1 is a tumor suppressor gene that inhibits the transmembrane receptor Smoothened (SMO) and prevents downstream Hh signaling pathway activation^{85,86}. The Hh pathway is activated when one of the Hh ligands including Sonic hedgehog (SHh), Indian hedgehog (IHh), or Desert Hedgehog (DHh) bind to PTCH1 and disrupt SMO inhibition⁸⁶. Inactivating mutations in PTCH1 lead to ligand-independent signaling of Hh, as PTCH1 no longer prevents SMO activity⁸⁶. Germline mutations in PTCH1 are associated with basal cell nevus syndrome (BCNS) or Gorlin Syndrome with a predisposition to non-cancerous and cancerous tumors including basal cell carcinoma^{86,87}.

Alterations and prevalence: Inactivating mutations in PTCH1 are observed in 85% of sporadic basal cell carcinomas⁸⁷. Somatic mutations in PTCH1 are also observed in 11% of uterine corpus endometrial carcinoma and 4-5% of stomach adenocarcinoma, skin cutaneous melanoma, cholangiocarcinoma, esophagus adenocarcinoma, colorectal adenocarcinoma, and mesothelioma^{5,6}.

Potential relevance: Currently, no therapies are approved for PTCH1 aberrations.

SETD2 (SET domain containing 2, histone lysine methyltransferase)

Background: The SETD2 gene encodes the SET domain containing 2 histone lysine methyltransferase, a protein responsible for the trimethylation of lysine-36 on histone H3 (H3K36)^{88,89}. Methylation of H3K36 is a hallmark of active transcription and can be either mono-, di-, or tri-methylated where di- and tri-methylation are thought to be responsible for transcriptional regulation⁹⁰. Trimethylation of H3K36 by SETD2 promotes post-transcriptional gene silencing and prevents aberrant transcriptional initiation^{91,92}. SETD2 trimethylation activity is also observed to be involved in DNA repair through the recruitment of DNA repair machinery⁸⁹. Specifically, H3K36 tri-methylation by SETD2 has been shown to regulate mismatch repair (MMR) in vivo, wherein the loss of SETD2 results in MMR deficiency (dMMR) and consequent microsatellite instability (MSI)⁹³. Both copy number deletion and mutations resulting in SETD2 loss of function have been observed in a variety of cancers, suggesting a tumor suppressor role for SETD2^{89,94}.

Alterations and prevalence: Inactivating somatic mutations in SETD2 were first described in clear cell renal cell carcinoma (ccRCC) and are observed to be predominantly missense or truncating^{5,94,95}. Mutations at codon R1625 are observed to be the most recurrent with R1625C having been identified to result in loss of SETD2 H3K36 trimethylase activity^{5,88}. SETD2 mutation is observed in about 14% of uterine cancer, 12% of ccRCC, 9% of mesothelioma, and 6-7% of melanoma, lung adenocarcinoma, papillary renal cell carcinoma (pRCC), colorectal and bladder cancers⁸⁸. Biallelic loss of SETD2 is observed in about 6% of diffuse large B-cell lymphoma, and about 3% of ccRCC and mesothelioma⁸⁸.

Potential relevance: Currently, no therapies are approved for SETD2 aberrations. Mutations in SETD2 can be used to support diagnosis of hepatosplenic T-cell lymphoma (HSTCL)⁹⁶.

Biomarker Descriptions (continued)

SMARCA4 (SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily a, member 4)

Background: The SMARCA4 gene encodes the SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily A, member 4 protein¹. SMARCA4, also known as BRG1, is a core member of ATP-dependent, multisubunit SWI/SNF chromatin-remodeling complex, along with SMARCB1/SNF5, SMARCC1/BAF155, SMARCC2/BAF170, and SMARCA2/BRM³. The SWI/SNF complex remodels chromatin at promoter and enhancer elements to alter and regulate gene expression^{3,4}. SMARCA4 and SMARCA2 are highly homologous and are mutually exclusive ATPase catalytic subunits for SWI/SNF chromatin remodeling complexes^{3,4}. Germline loss of function mutations in SMARCA4 are associated with atypical teratoid/rhabdoid tumors (AT/RT), and a rare form of ovarian cancer called small cell carcinoma of the ovary, hypercalcemic type (SCCOHT), which highlights the tumor suppressor function of SMARCA4.^{97,98}

Alterations and prevalence: Mutations in SWI/SNF complex subunits are the most commonly mutated chromatin modulators in cancer and have been observed in 20% of all tumors⁴. Recurrent somatic mutations in SMARCA4 are observed in 10% of skin cutaneous melanoma and uterine corpus endometrial carcinoma, and 7% of esophageal adenocarcinoma^{5,6}.

Potential relevance: Currently, no therapies are approved for SMARCA4 aberrations. SMARCA4 mutations and deletions are considered a diagnostic marker for the SMARCA4-deficient uterine sarcoma (SDUS) subtype⁹⁹.

TP53 (tumor protein p53)

Background: The TP53 gene encodes the p53 tumor suppressor protein that binds to DNA and activates transcription in response to diverse cellular stresses to induce cell cycle arrest, apoptosis, or DNA repair. In unstressed cells, TP53 is kept inactive by targeted degradation via MDM2, a substrate recognition factor for ubiquitin-dependent proteolysis. Alterations in TP53 is required for oncogenesis as they result in loss of protein function and gain of transforming potential¹⁰⁰. Germline mutations in TP53 are the underlying cause of Li-Fraumeni syndrome, a complex hereditary cancer predisposition disorder associated with early-onset cancers^{101,102}.

Alterations and prevalence: TP53 is the most frequently mutated gene in the cancer genome with approximately half of all cancers experiencing TP53 mutations. Ovarian, head and neck, esophageal, and lung squamous cancers have particularly high TP53 mutation rates (60-90%)^{5,6,15,103,104,105}. Approximately two-thirds of TP53 mutations are missense mutations and several recurrent missense mutations are common including substitutions at codons R158, R175, Y220, R248, R273, and R282^{5,6}. Invariably, recurrent missense mutations in TP53 inactivate its ability to bind DNA and activate transcription of target genes^{106,107,108,109}.

Potential relevance: The small molecule p53 reactivator, PC14586, received a fast track designation (2020) by the FDA for advanced tumors harboring a TP53 Y220C mutation¹¹⁰. The FDA has granted fast track designation (2019) to the p53 reactivator, eprenetapopt,¹¹¹ and breakthrough designation¹¹² (2020) in combination with azacitidine or azacitidine and venetoclax for acute myeloid leukemia patients (AML) and myelodysplastic syndrome (MDS) harboring a TP53 mutation, respectively. In addition to investigational therapies aimed at restoring wild-type TP53 activity, compounds that induce synthetic lethality are also under clinical evaluation^{113,114}. TP53 mutations confer poor prognosis in multiple blood cancers including AML, MDS, myeloproliferative neoplasms (MPN), and chronic lymphocytic leukemia (CLL)^{69,70,115,116,117}. In mantle cell lymphoma, TP53 mutations are associated with poor prognosis when treated with conventional therapy including hematopoietic cell transplant¹¹⁸. Mono- and bi-allelic mutations in TP53 confer unique characteristics in MDS, with multi-hit patients also experiencing associations with complex karyotype, few co-occurring mutations, and high-risk disease presentation as well as predicted death and leukemic transformation independent of the IPSS-R staging system¹¹⁹.

TSC2 (TSC complex subunit 2)

Background: The TSC2 gene encodes the tuberin protein. TSC2 and TSC1 (also known as hamartin) form a complex through their respective coiled-coil domains¹²⁰. The TSC1-TSC2 complex is a negative regulator of the mTOR signaling pathway that regulates cell growth, cell proliferation, and protein and lipid synthesis¹²¹. Specifically, the TSC1-TSC2 complex acts as a GTPase activating (GAP) protein that inhibits the G-protein RHEB and keeps it in an inactivated state (RHEB-GDP). GTP bound RHEB (RHEB-GTP) is required to activate the mTOR complex 1 (mTORC1). TSC1 and TSC2 are tumor suppressor genes. Loss of function mutations in TSC1 and TSC2 lead to dysregulation of the mTOR pathway^{120,122}. Inactivating germline mutations in TSC1 and TSC2 are associated with tuberous sclerosis complex (TSC), an autosomal dominant neurocutaneous and progressive disorder that presents with multiple benign tumors in different organs¹²⁰.

Biomarker Descriptions (continued)

Alterations and prevalence: Somatic mutations are observed in up to 8% of skin cutaneous melanoma, 7% of uterine corpus endometrial carcinoma, and 4% of cervical squamous cell carcinoma^{5,6}.

Potential relevance: Currently, no therapies are approved for TSC2 aberrations.

Relevant Therapy Summary

☒ In this cancer type ☐ In other cancer type ☒ In this cancer type and other cancer types ☒ No evidence

PALB2 G1135A

Relevant Therapy	FDA	NCCN	Clinical Trials*
olaparib	☐	☐	☒ (II)
rucaparib	☒	☐	☒
dostarlimab, niraparib	☒	☒	☒ (II)
niraparib	☒	☒	☒ (II)
pamiparib, tislelizumab	☒	☒	☒ (II)
talazoparib	☒	☒	☒ (II)
talazoparib, atezolizumab + talazoparib	☒	☒	☒ (II)
ZEN-3694, talazoparib	☒	☒	☒ (II)
sacituzumab govitecan, berzosertib	☒	☒	☒ (I/II)
senaparib, IMP-9064	☒	☒	☒ (I/II)
cabozantinib, pamiparib	☒	☒	☒ (I)
copanlisib, olaparib, durvalumab	☒	☒	☒ (I)
HH-CYH33, olaparib	☒	☒	☒ (I)

BRAF R671Q

Relevant Therapy	FDA	NCCN	Clinical Trials*
trametinib	☒	☐	☒
abemaciclib + temuterkib	☒	☒	☒ (II)
binimetinib, encorafenib	☒	☒	☒ (II)
regorafenib	☒	☒	☒ (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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Relevant Therapy Summary (continued)

In this cancer type

In other cancer type

In this cancer type and other cancer types

No evidence

Relevant Therapy	FDA	NCCN	Clinical Trials*
sorafenib + trametinib	×	×	● (II)
vemurafenib, dabrafenib	×	×	● (II)
ASTX029	×	×	● (I/II)
GH-55	×	×	● (I/II)
mirdametinib, BGB-3245	×	×	● (I/II)
mirdametinib, lifirafenib	×	×	● (I/II)
OKI-179, binimetinib	×	×	● (I/II)
ABM-1310, cobimetinib	×	×	● (I)
AZD-0364, nivolumab	×	×	● (I)
BBP-398	×	×	● (I)
cobimetinib, belvarafenib	×	×	● (I)
ET0038	×	×	● (I)
JSI-1187	×	×	● (I)
JZP-815	×	×	● (I)
QLH11906	×	×	● (I)

Relevant Therapy	FDA	NCCN	Clinical Trials*
dostarlimab, niraparib	×	×	● (II)
nivolumab	×	×	● (II)
pamiparib, tislelizumab	×	×	● (II)
talazoparib	×	×	● (II)
CCS-1477	×	×	● (I/II)
CPI-0209	×	×	● (I/II)
senaparib, IMP-9064	×	×	● (I/II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type

☐ In other cancer type

☒ In this cancer type and other cancer types

☒ No evidence

ARID1A Q581* (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
cabozantinib, pamiparib	×	×	● (I)
M-1774	×	×	● (I)

SMARCA4 P22R

Relevant Therapy	FDA	NCCN	Clinical Trials*
dostarlimab, niraparib	×	×	● (II)
nivolumab, ipilimumab	×	×	● (II)
atezolizumab, tiragolumab	×	×	● (I/II)
tazemetostat, nivolumab, ipilimumab	×	×	● (I/II)

TP53 R248W

Relevant Therapy	FDA	NCCN	Clinical Trials*
niraparib	×	×	● (II)
olaparib	×	×	● (II)

SETD2 S188*

Relevant Therapy	FDA	NCCN	Clinical Trials*
niraparib	×	×	● (II)
olaparib	×	×	● (II)

PTCH1 Q417E

Relevant Therapy	FDA	NCCN	Clinical Trials*
taladegib	×	×	● (II)
vismodegib	×	×	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

In this cancer type

In other cancer type

In this cancer type and other cancer types

No evidence

TSC2 E588Q

Relevant Therapy	FDA	NCCN	Clinical Trials*
temsirolimus	×	×	(II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

PALB2 G1135A

NCT ID	Title	Phase
NCT04779151	An Open Label Phase II Basket Trial Exploring The Efficacy And Safety Of The Combination Of Niraparib And Dostarlimab In Patients With DNA Repair-Deficient Or Platinum-Sensitive Solid Tumors	II
NCT03375307	A Phase II Study of Olaparib (AZD2281) in Patients With Metastatic/Advanced Urothelial Carcinoma and Other Genitourinary Tumors With DNA-Repair Defects	II
NCT04992013	Genomically Guided Phase II Study to Evaluate the Clinical Benefit of Niraparib in Tumors Metastatic to the CNS	II
NCT05169437	A Single-Arm Phase-II Study of Niraparib in Locally Advanced or Metastatic Solid Tumor Patients With PALB2 Mutations	II
NCT02029001	A Two-period, Multicenter, Randomized, Open-label, Phase II Study Evaluating the Clinical Benefit of a Maintenance Treatment Targeting Tumor Molecular Alterations in Patients with Progressive Locally-advanced or Metastatic Solid Tumors.	II
NCT03742895	A Phase II Study of Olaparib Monotherapy in Participants With Previously Treated, Homologous Recombination Repair Mutation (HRRm) or Homologous Recombination Deficiency (HRD) Positive Advanced Cancer	II
NCT03967938	Efficacy of Olaparib in Advanced Cancers Occurring in Patients With Germline Mutations or Somatic Tumor Mutations in Homologous Recombination Genes	II
NCT04985721	An Open Label, Signal Seeking, Translational, Phase II Trial of Pamiparib in Combination With Tislelizumab in Patients With Advanced Tumours With Homologous Recombination Repair Defects	II
NCT04550494	A Pharmacodynamics-Driven Trial of Talazoparib, an Oral PARP Inhibitor, in Patients With Advanced Solid Tumors and Aberrations in Genes Involved in DNA Damage Response	II
NCT05097599	Strata PATH™ (Precision Indications for Approved Therapies): A Study Evaluating the Clinical Activity and Safety of Approved Drugs Within Biomarker-Guided Patients With Solid Tumors	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II
NCT05327010	Phase II Trial of the Combination of the BET Inhibitor, ZEN003694 (ZEN-3694), and the PARP Inhibitor Talazoparib, in Patients With Molecularly-Selected Solid Tumors (CombET)	II

Clinical Trials Summary (continued)

PALB2 G1135A (continued)

NCT ID	Title	Phase
NCT04826341	A Phase I/II Study of Sacituzumab Govitecan Plus Berzosertib in Small Cell Lung Cancer, Extra-Pulmonary Small Cell Neuroendocrine and Homologous Recombination-Deficient Cancers Resistant to PARP Inhibitors	I/II
NCT05269316	A First-in-human, Phase I/II, Open-label, Multi-center, Dose-escalation and Dose-expansion Study to Evaluate Safety, Tolerability, Pharmacokinetics, and Anti-tumor Activity of the ATR Inhibitor IMP9064 Monotherapy and in Combination With PARP Inhibitor Senaparib in Patients With Advanced Solid Tumors	I/II
NCT05038839	A Phase I Study of Cabozantinib and Pamiparib to Evaluate Triple Inhibition of PARP, VEGFR and c-MET in Advanced Homologous Recombination Deficient Malignancies	I
NCT03842228	A Phase Ib Biomarker-Driven Combination Trial of Copanlisib, Olaparib, and Durvalumab (MEDI4736) in Patients With Advanced Solid Tumors	I
NCT04586335	Open Label, Phase Ib Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Clinical Activity of CYH33, an Oral α -specific PI3K Inhibitor in Combination With Olaparib, an Oral PARP Inhibitor in Patients With Advanced Solid Tumors.	I

BRAF R671Q

NCT ID	Title	Phase
NCT04534283	A Phase II Basket Trial of an ERK1/2 Inhibitor (LY3214996) in Combination With Abemaciclib for Patients Whose Tumors Harbor Pathogenic Alterations in BRAF, RAF1, MEK1/2, ERK1/2, and NF1	II
NCT03839342	Binimetinib and Encorafenib for the Treatment of Advanced Solid Tumors With Non-V600E BRAF Mutations	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II
No NCT ID	Cross-Institutional, Prospective, Open Label, Single Group Basket Study of Combined CRAF and MEK Inhibition in Advanced-Stage Malignancies Harboring BRAF Mutations with Impaired Kinase Activity	II
NCT03239015	Efficacy and Safety of Precision Therapy in Refractory Tumor (Long March Pathway)	II
NCT04190628	A Phase I, First-In-Human, Multicenter, Open-Label Dose Escalation and Dose Expansion Study of ABM-1310, as a Monotherapy and a Combination Therapy, Administered Orally in Adult Patients With Advanced Solid Tumors Harboring BRAF Mutations	I
NCT03905148	A Phase Ib, Open-Label, Dose-escalation and Expansion Study to Investigate the Safety, Pharmacokinetics and Antitumor Activities of a RAF Dimer Inhibitor BGB-283 in Combination With MEK Inhibitor PD-0325901 in Patients With Advanced or Refractory Solid Tumors	I/II
NCT03284502	A Phase Ib, Open-label, Multicenter, Dose Escalation Study of the Safety, Tolerability, and Pharmacokinetics of HM95573 in Combination With Either Cobimetinib or Cetuximab in Patients With Locally Advanced or Metastatic Solid Tumors	I
No NCT ID	A Phase I/II Clinical Study to Evaluate the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics Characteristics and Efficacy of Oral Administration of GH55 in Patients with Advanced Solid Tumors with Mutations in MAPK Signaling Pathway	I/II

Clinical Trials Summary (continued)

BRAF R671Q (continued)

NCT ID	Title	Phase
NCT05340621	NAUTILUS: A Phase I b/II Study of OKI-179 Plus Binimetinib in Patients With Advanced Solid Tumors and Activating Mutations in the RAS Pathway (Phase 1b) and in Patients With Advanced NRAS-Mutated Melanoma (Phase 2)	I/II
NCT05557045	Phase I, FIH, Open-label, Nonrandomized, Multicenter Study of JZP815 in Participants With Advanced or Metastatic Solid Tumors Harboring Alterations in the MAPK Pathway	I
NCT03520075	A Phase I/II Study of the Safety, Pharmacokinetics, and Activity of ASTX029 in Subjects With Advanced Solid Tumors	I/II
NCT05580770	A Phase I/IIa Open-Label, Dose Escalation and Expansion Study to Investigate the Safety, Pharmacokinetics, Pharmacodynamics and Efficacy of Mirdametinib in Combination With BGB-3245 in Patients With Advanced Solid Tumors	I/II
NCT04305249	A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of ATG-017 Monotherapy or Combination Therapy With Nivolumab in Patients With Advanced Solid Tumors and Hematological Malignancies	I
NCT04528836	A Phase I/IB First-in-Human Study of the SHP2 Inhibitor BBP-398 (Formerly Known as IACS-15509) in Patients With Advanced Solid Tumors	I
NCT05354843	A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of SHP2 Inhibitor ET0038 Monotherapy in Patients With Advanced Solid Tumors	I
NCT04418167	A Phase I Study of ERK1/2 Inhibitor JSI-1187 Administered as Monotherapy and in Combination with Dabrafenib for the Treatment of Advanced Solid Tumors with MAPK Pathway Mutations	I
NCT05488821	A Phase I Clinical Study to Evaluate the Safety, Tolerability and Pharmacokinetics of the Oral Pan-RAF Inhibitor QLH11906 in Subjects With Advanced Solid Tumors Harboring MAPK Pathway Alterations.	I

ARID1A Q581*

NCT ID	Title	Phase
NCT04779151	An Open Label Phase II Basket Trial Exploring The Efficacy And Safety Of The Combination Of Niraparib And Dostarlimab In Patients With DNA Repair-Deficient Or Platinum-Sensitive Solid Tumors	II
NCT04104776	A Phase I/II Study of CPI-0209 in Patients With Advanced Solid Tumors and Lymphomas	I/II
NCT04957615	A Phase II Clinical Trial to Investigate ARID1A Mutation and CXCL13 Expression in the Pre-Treatment Tumor Samples as a Combinatorial Predictive Biomarker for Immune Checkpoint Therapy in Metastatic Solid Tumors	II
NCT04985721	An Open Label, Signal Seeking, Translational, Phase II Trial of Pamiparib in Combination With Tislezumab in Patients With Advanced Tumours With Homologous Recombination Repair Defects	II
NCT04550494	A Pharmacodynamics-Driven Trial of Talazoparib, an Oral PARP Inhibitor, in Patients With Advanced Solid Tumors and Aberrations in Genes Involved in DNA Damage Response	II
NCT03568656	An Open-label Phase I/IIa Study to Evaluate the Safety and Efficacy of CCS1477 as Monotherapy and in Combination, in Patients With Advanced Solid/Metastatic Tumours.	I/II

Clinical Trials Summary (continued)

ARID1A Q581* (continued)

NCT ID	Title	Phase
NCT05269316	A First-in-human, Phase I/II, Open-label, Multi-center, Dose-escalation and Dose-expansion Study to Evaluate Safety, Tolerability, Pharmacokinetics, and Anti-tumor Activity of the ATR Inhibitor IMP9064 Monotherapy and in Combination With PARP Inhibitor Senaparib in Patients With Advanced Solid Tumors	I/II
NCT05038839	A Phase I Study of Cabozantinib and Pamiparib to Evaluate Triple Inhibition of PARP, VEGFR and c-MET in Advanced Homologous Recombination Deficient Malignancies	I
NCT04170153	An Open-label, Multicenter Trial of the Safety, Tolerability, and Pharmacokinetic/Pharmacodynamic Profile of M1774 in Participants With Metastatic or Locally Advanced Unresectable Solid Tumors (DDRiver Solid Tumors 301)	I

SMARCA4 P22R

NCT ID	Title	Phase
NCT04779151	An Open Label Phase II Basket Trial Exploring The Efficacy And Safety Of The Combination Of Niraparib And Dostarlimab In Patients With DNA Repair-Deficient Or Platinum-Sensitive Solid Tumors	II
NCT04416568	Phase II Proof of Concept Study of Nivolumab and Ipilimumab in Children and Young Adults With Relapsed or Refractory INI1-negative Cancers	II
NCT05286801	A Phase I/II Study of Tiragolumab (NSC# 827799) and Atezolizumab (NSC# 783608) in Patients With Relapsed or Refractory SMARCB1 or SMARCA4 Deficient Tumors	I/II
NCT05407441	TAZNI: A Phase I/II Combination Trial of Tazemetostat With Nivolumab and Ipilimumab for Children With INI1-Negative or SMARCA4-Deficient Tumors	I/II

TP53 R248W

NCT ID	Title	Phase
NCT03375307	A Phase II Study of Olaparib (AZD2281) in Patients With Metastatic/Advanced Urothelial Carcinoma and Other Genitourinary Tumors With DNA-Repair Defects	II
NCT02029001	A Two-period, Multicenter, Randomized, Open-label, Phase II Study Evaluating the Clinical Benefit of a Maintenance Treatment Targeting Tumor Molecular Alterations in Patients with Progressive Locally-advanced or Metastatic Solid Tumors.	II
NCT04992013	Genomically Guided Phase II Study to Evaluate the Clinical Benefit of Niraparib in Tumors Metastatic to the CNS	II

Clinical Trials Summary (continued)

SETD2 S188*

NCT ID	Title	Phase
NCT03375307	A Phase II Study of Olaparib (AZD2281) in Patients With Metastatic/Advanced Urothelial Carcinoma and Other Genitourinary Tumors With DNA-Repair Defects	II
NCT04992013	Genomically Guided Phase II Study to Evaluate the Clinical Benefit of Niraparib in Tumors Metastatic to the CNS	II

PTCH1 Q417E

NCT ID	Title	Phase
NCT05199584	A Phase II, Multi-Center Study Evaluating the Safety and Efficacy of ENV-101 (Taladegib) in Patients With Advanced Solid Tumors Harboring PTCH1 Loss of Function Mutations	II
No NCT ID	Single Arm, Open Label, Signal Seeking, Phase IIa Trial of The Activity of Vismodegib in Patients With Tumours Harboursing PTCH1 or SMO Mutations	II

TSC2 E588Q

NCT ID	Title	Phase
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II

Assay Information

Methodology, Scope, and Application:The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity:The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations:This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient’s history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient’s) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal

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(i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent OncoPrint Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OncoPrint Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trial, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGF, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSPO2, RSPO3, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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